



資料結構

Data Structure

加分題 2

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Q1: Valid sudoku

檢查數讀是否符合規定

Code

```
#include <vector>
using namespace std;
class Solution{
public:
    bool isValidSudoku(vector<vector<char>>&board){
        // 檢查列
        for(int i=0;i<9;i++){
            bool seen[10]={false};
            for(int j=0;j<9;j++){
                if(board[i][j]!='.'){// 只檢查數字
                    int num=board[i][j]-'0';
                    if(seen[num])return false;// 有重複出現
                    seen[num]=true;// 沒有重複出現
                }
            }
        }
        // 檢查行
        for(int j=0;j<9;j++){
            bool seen[10]={false};
            for(int i=0;i<9;i++){
                if(board[i][j]!='.'){
                    int num=board[i][j]-'0';
                    if(seen[num])return false;
                    seen[num]=true;
                }
            }
        }
        // 檢查 9 宮格
        for(int row=0;row<9;row+=3){
            for(int col=0;col<9;col+=3){
                bool seen[10]={false};
                for(int i=row;i<row+3;i++){
                    for(int j=col;j<col+3;j++){
                        if(board[i][j]!='.'){
```

```

        int num=board[i][j]-'0';
        if(seen[num])return false;
        seen[num]=true;
    }
}
}
}
}
return true;
}
};

```

Result

The screenshot displays a coding platform interface with the following components:

- Submission Status:** Accepted, 507 / 507 testcases passed. Submitted by Liao_Yanxuan on Jun 03, 2026 at 23:37.
- Performance Metrics:**
 - Runtime: 3 ms, Beats 54.60%.
 - Memory: 21.67 MB, Beats 86.40%.
- Runtime Graph:** A bar chart showing the distribution of runtime times across various test cases, with most cases falling between 2ms and 14ms.
- Code Editor:** Contains the C++ code for the `isValidSudoku` function. The code uses a `seen` array to track digits and checks for duplicates in rows and columns.


```

1 #include <vector>
2 using namespace std;
3 class Solution{
4 public:
5     bool isValidSudoku(vector<vector<char>>&board){
6         //檢查列
7         for(int i=0;i<9;i++){
8             bool seen[10]={false};
9             for(int j=0;j<9;j++){
10                 if(board[i][j]!='.'){//只檢查數字
11                     int num=board[i][j]-'0';
12                     if(seen[num])return false;//有重複出現
13                     seen[num]=true;//沒有重複出現
14                 }
15             }
16         }
17         //檢查行
18         for(int j=0;j<9;j++){
19             bool seen[10]={false};
20             for(int i=0;i<9;i++){
21                 if(board[i][j]!='.'){

```
- Test Results:** Shows 'Accepted' with a runtime of 0 ms for the test cases.
- System Bar:** At the bottom, it shows the Windows taskbar with the search bar, task view, and system tray indicating 30°C and the time 12:35 on 2026/6/4.